Backtracking-weighted PageRank is not coordinatewise monotone: minimal counterexamples, a certified census and endpoint tests

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Abstract

Plain-language summary. PageRank scores the vertices of a network by a random walk that occasionally restarts. A walk may be told to avoid stepping straight back along the edge it has just used (non-backtracking), to behave as usual, or to bounce back and forth. Glover, Jones, Kempton and Oveson conjectured that, as the return weight μ is turned up, every vertex score moves in one direction only. If true, two cheap calculations would tell an analyst which way the expensive non-backtracking score lies. We show that the conjecture is false, and that its failure is common rather than exotic.

Results. On the five-vertex house graph the roof vertex first loses and then regains score; an exact rational certificate proves this, and a closed cubic formula classifies every damping value. Among graphs of minimum degree two, five vertices is the smallest order for *every* damping value: the house covers $\varepsilon > \varepsilon_c \approx 0.2407$ and the gem covers $\varepsilon \leq 1/3$. If leaves are allowed, the four-vertex paw fails for small and for large damping, where four is then the smallest order; at intermediate damping it is five. Even a five-vertex tree fails. An exact census of all 995 connected graphs on two to seven vertices, at five damping values, finds a nonmonotone vertex in 675 of the 853 seven-vertex graphs. Both readings of the conjecture's spread clause also fail exactly, even when the extreme vertices are unique. We prove a two-step criterion that explains which vertices dip at small damping, and an exact formula for the slope at ordinary PageRank that gives cheap sufficient tests. Numerical experiments find that 10–41% of the vertices of standard networks are nonmonotone; the excursions are small and rankings rarely change. We also characterise exact invariance under backtracking and refute the degree-tail Conjecture 3.2 of the same paper, in its local-weak-convergence reading, for every tail index above one. Every theorem is backed by an exact proof; every computational claim is labelled as exact or numerical.

Contents

1 Introduction **2**
1.1 The question . . . 2
1.2 A three-number certificate . . . 2
1.3 Roadmap and summary of results . . . 3

2 The model **3**

3 An exact response on the house graph **4**

4 The complete damping classification of the roof vertex **6**

5 Smallest counterexamples and large examples **7**

6 Why vertices dip, and cheap tests **9**
6.1 A two-step picture 9
6.2 The slope at ordinary PageRank 10

7 A certified census on at most seven vertices **10**

8 The spread clause **11**

9 Exact invariance **12**

10 Prevalence and practical impact (numerical) **13**

11 Degree tails do not determine infinite-backtracking tails **14**

12 Verification, literature boundary and scope **15**

13 Conclusions and open problems **15**

1 Introduction

1.1 The question

PageRank combines movement along edges with occasional restart. A walk with memory can change the weight of an immediate return along the edge just used. The parameter $\mu \geq 0$ gives that return weight μ and each alternative weight one. Thus $\mu = 0$ forbids immediate returns, $\mu = 1$ is ordinary PageRank, and $\mu \to \infty$ makes the walk oscillate on a single edge. Glover, Jones, Kempton and Oveson [1, 2] studied the resulting vertex scores π_μ and stated two conjectures in Sections 3.3 and 3.4 of their paper. We quote them verbatim from arXiv:2211.13353v2 (9 February 2026). The journal version [1] was not accessible to us, so we could not compare its wording.

Conjecture 1 ([2, Conjecture 3.1])**.** "The function $(\pi_\mu)_i$ is monotonic for all i and

$$\max_i \pi_\mu(i) - \min_i \pi_\mu(i) \leq \max_i \pi_{\mu+\alpha}(i) - \min_i \pi_{\mu+\alpha}(i)$$

for all $\alpha > 0$."

Conjecture 2 ([2, Conjecture 3.2])**.** "If $d_i \sim X$ where X is a regularly-varying random variable and d_i is the degree of node $i \in V(G)$, then $\pi_\infty(i)$ will also be scale-free."

Conjecture 1 is a conjunction of a *coordinatewise clause* and a *spread clause*. The coordinatewise clause is the substantive one. The source motivates it computationally: the infinite-backtracking score π_∞ has a cheap closed form, so if each coordinate were monotone, comparing π_∞ with ordinary PageRank π_1 would tell which side of π_1 the non-backtracking score π_0 lies on.

1.2 A three-number certificate

For the house graph (Figure 1) with uniform restart and damping $\varepsilon = 1/2$, the roof vertex has

$$\pi_0(4) = \frac{638}{3615}, \qquad \pi_1(4) = \frac{22}{125}, \qquad \pi_\infty(4) = \frac{8}{45}, \tag{1}$$

so that $\pi_0(4) - \pi_1(4) = 44/90375 > 0$ and $\pi_\infty(4) - \pi_1(4) = 2/1125 > 0$. Both endpoints lie strictly above the middle value, so the roof score is not monotone in μ. Monotone interpolation would put the two endpoints on opposite sides of $\pi_1(4)$, so the proposed shortcut points the wrong way here.

1.3 Roadmap and summary of results

Section 2 fixes the model, including leaves. Sections 3–4 give the exact house response and its damping classification. Section 5 identifies the smallest counterexamples, with and without the minimum-degree assumption, and lifts them to arbitrarily large graphs. Section 6 explains the dip and gives cheap tests. Section 7 reports the certified census. Section 8 settles the spread clause. Section 9 characterises exact invariance. Section 10 measures prevalence and practical impact numerically. Section 11 treats Conjecture 2. Section 12 describes the verification code, and Section 13 concludes. Table 1 lists the results. "Proved" means a complete mathematical proof in this paper (some steps are exact polynomial identities checked by the accompanying code). "Certified" means an exact computation: a finite exact certificate, or an exhaustive exact search where so stated. "Numerical" means floating-point evidence only.

Table 1: Summary of results. Status: P = proved; C = certified exact computation; N = numerical. Novelty: "new" means not found within our literature search (Section 12); "inherited" means taken from, or a direct adaptation of, the cited work.

Result	Content	Status	Novelty
Thms 3.1, 4.1	House roof: cubic formula; monotone iff $\varepsilon \le \varepsilon_c$; one minimum, at $\mu = 1$ iff $\varepsilon = 4/9$	P	new
Prop. 4.3	Endpoint shortcut is wrong for the roof iff $\varepsilon > \varepsilon_0 \approx 0.3291$	P	new
Thm 5.2	Min. degree ≥ 2: order 5 is minimal for *every* ε (house $\cup$ gem)	P	new
Prop. 5.3, Cor. 5.4	With leaves: the paw fails iff $\varepsilon \in (0, \varepsilon_1) \cup (\varepsilon_2, 1)$; minimal order is 4 there and 5 on $[\varepsilon_1, \varepsilon_2]$	P	new
Prop. 5.5	A five-vertex tree with a nonmonotone leaf on $(0, 1)$	C	new
Lemma 5.6, Cor. 5.7	Failures lift through covers to every order $5k$	P	lemma inherited [8]
Thm 6.1	Small-damping two-step criterion; sign test σ_j	P	new
Prop. 6.2, Cor. 6.3	Exact slope at $\mu = 1$; three endpoint tests	P	new
Section 7	Census of all connected graphs on ≤ 7 vertices, five ε	C	new
Prop. 8.1, 8.2	Spread clause fails as printed and when reversed, also with unique extreme vertices	P, C	new
Thm 9.1	Exact invariance iff $P^2 v = v$	P	converse new; sufficiency [5, 1]
Section 10	Prevalence, dip sizes, shortcut failures, rank stability	N	new
Prop. 10.1	An exact transient rank reversal (Les Misérables)	C	new
Thm 11.2	Conjecture 2 fails for every tail index $\alpha > 1$	P	mechanism inherited [17]
Prop. 11.1	$n\pi_\infty(i) \le (1 + \varepsilon d_i/2)/(1 + \varepsilon)$	P	elementary

2 The model

Let $G = (V, E)$ be a finite connected simple undirected graph with degrees d_i, and let $\vec{E}$ be the set of the two orientations of each edge. A state $(i, j) \in \vec{E}$ records the directed edge last used; we call i its *tail*. For $\mu \ge 0$ define the column-stochastic matrix Q_μ by

$$(Q_\mu)_{(j,k),(i,j)} = \frac{1 + (\mu - 1)\mathbf{1}_{\{k=i\}}}{d_j - 1 + \mu} \quad (k \sim j), \tag{2}$$

with all other entries zero. If $d_j \ge 2$, the denominator satisfies $d_j - 1 + \mu \ge 1$.

Leaves. If $d_j = 1$, the only continuation is the return, and (2) gives it weight $\mu/\mu = 1$ for every $\mu > 0$. At $\mu = 0$ the source's kernel is undefined at a leaf, because the corresponding entry of

$\hat{D} - I$ vanishes. We use the same forced return at $\mu = 0$, which is the limit $\mu \to 0^+$, that is, the continuous extension of the source kernel; this endpoint choice is ours. For $\mu > 0$ the kernel agrees with node2vec with return parameter $p = 1/\mu$ and in–out parameter $q = 1$ [14, 21]. It differs from some of the leaf conventions of Aleja et al. [3]. Every nonmonotonicity statement we make on a graph with leaves concerns the open range $\mu \in (0, \infty)$, where no convention is needed. For graphs of minimum degree at least two, which is the setting of the source's theorems, the kernel needs no convention at any $\mu \geq 0$.

Let v be a probability vector on V, spread restart mass evenly over outgoing edges, $u_{ij} = v_i/d_i$, and fix a damping (follow) probability $0 < \varepsilon < 1$. The edge PageRank and vertex scores are

$$q_\mu = (1 - \varepsilon)u + \varepsilon Q_\mu q_\mu, \qquad \pi_\mu(i) = \sum_{j \sim i} q_\mu(i, j). \tag{3}$$

Since $\|\varepsilon Q_\mu\|_1 = \varepsilon < 1$, the solution $q_\mu = (1 - \varepsilon) \sum_{t \geq 0} \varepsilon^t Q_\mu^t u$ is unique, nonnegative and of total mass one. For uniform restart $v_i = 1/n$ we say *uniform restart*. It is used throughout unless stated otherwise.

Remark 2.1 (Faithfulness and projection). In the notation of [2, Definition 2.4], $Q_\mu = C_\mu^T(\hat{D} - (1 - \mu)I)^{-1}$, $u = T^T(TT^T)^{-1}v$ and $\pi_\mu = T\hat{\pi}_\mu$; this is the tail projection. (The source writes $H_\mu^T \hat{y} = \hat{y}$ in some proofs, which is inconsistent with the column-stochastic H_μ of its Definition 2.4. We use the column-stochastic reading.) The head projection $H_\mu(j) = \sum_i q_\mu(i, j)$ satisfies $\pi_\mu = (1 - \varepsilon)v + \varepsilon H_\mu$, because every state leaving j is entered either by restart at j or from a state with head j. Hence monotonicity of a coordinate does not depend on which projection is used. On $\mu \in [0, 1]$ the model is the unweighted, leaf-free special case of [3, Definitions 3.5, 3.7]. Related formulations are [5, 10, 22]. It is a first-order memory (edge-lift, Hashimoto [16]) model; it should not be confused with multilinear or second-order PageRank [13, 7], whose fixed point is nonlinear at the vertex level.

The two endpoints have closed forms. With A_G the adjacency matrix, $D_G = \mathrm{diag}(d_i)$ and $P = A_G D_G^{-1}$,

$$\pi_1 = (1 - \varepsilon)(I - \varepsilon P)^{-1}v, \qquad \pi_\infty = \frac{v + \varepsilon P v}{1 + \varepsilon}. \tag{4}$$

The second identity is [2, Proposition 3.3]. At $\mu = 1$ one also has $q_1(i, j) = \pi_1(i)/d_i$.

3 An exact response on the house graph

Use vertices $0, \ldots, 4$ and edges

$$E = \{01, 03, 04, 12, 14, 23\}. \tag{5}$$

Vertices 0 and 1 have degree three; vertices 2, 3 and the roof 4 have degree two.

Theorem 3.1 (Cubic roof response)**.** *Put $s = (1 - \varepsilon)\mu$. For the house graph with uniform restart,*

$$\pi_\mu(4) = \frac{N_\varepsilon(s)}{15 D_\varepsilon(s)}, \qquad N_\varepsilon(s) = \sum_{j=0}^{3} n_j s^j, \quad D_\varepsilon(s) = \sum_{j=0}^{3} d_j s^j, \tag{6}$$

with the coefficients in Table 2. In particular $D_\varepsilon(s) > 0$ for $s \geq 0$. At $\varepsilon = 1/2$, $\pi_\mu(4) = 2(12\mu^3 + 115\mu^2 + 346\mu + 319)/[15(9\mu^3 + 87\mu^2 + 263\mu + 241)]$.

Proof. The automorphism exchanging $0 \leftrightarrow 1$ and $2 \leftrightarrow 3$ preserves the kernel and the restart, so by uniqueness the edge masses are constant on its six orbits:

$$a = q_{01} = q_{10},\ b = q_{03} = q_{12},\ c = q_{30} = q_{21},\ d = q_{23} = q_{32},\ h = q_{04} = q_{14},\ f = q_{40} = q_{41}.$$

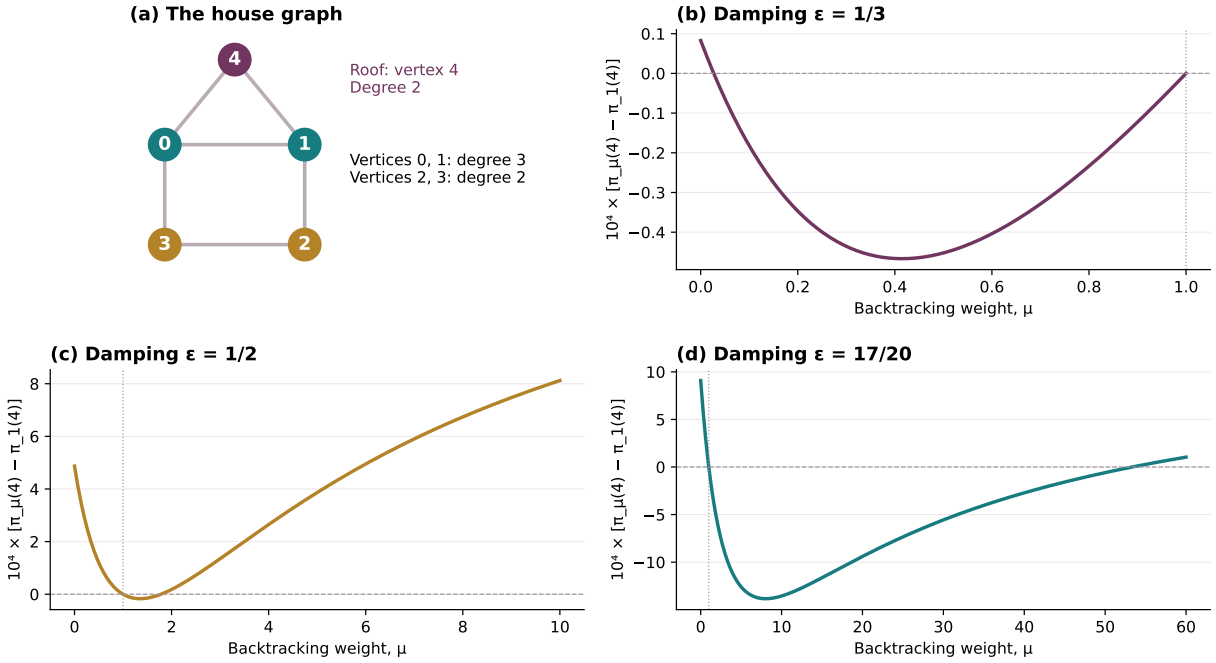


Figure 1: (a) The house graph. (b)–(d) The exact roof response (6), plotted as $10^4[\pi_\mu(4) - \pi_1(4)]$ for $\varepsilon = 1/3$ on $\mu \in [0, 1]$, $\varepsilon = 1/2$ on $[0, 10]$ and $\varepsilon = 17/20$ on $[0, 60]$. In (b) the minimum lies inside $[0, 1]$ because $\varepsilon < 4/9$. The curves illustrate the theorems; they are not part of any proof.

Table 2: Coefficients of the roof response (6). The variable $s = (1 - \varepsilon)\mu$ makes every coefficient a polynomial in ε with positive coefficients.

j	n_j	d_j
0	$2\varepsilon^5 + 4\varepsilon^4 + 9\varepsilon^3 + 10\varepsilon^2 + 8\varepsilon + 12$	$\varepsilon^5 + 2\varepsilon^4 + 3\varepsilon^3 + 4\varepsilon^2 + 4\varepsilon + 4$
1	$6\varepsilon^4 + 13\varepsilon^3 + 25\varepsilon^2 + 22\varepsilon + 24$	$3\varepsilon^4 + 6\varepsilon^3 + 10\varepsilon^2 + 10\varepsilon + 8$
2	$6\varepsilon^3 + 14\varepsilon^2 + 19\varepsilon + 15$	$3\varepsilon^3 + 6\varepsilon^2 + 8\varepsilon + 5$
3	$(1 + \varepsilon)(3 + 2\varepsilon)$	$(1 + \varepsilon)^2$

Write $r = \mu + 2$, $w = \mu + 1$, $t_A = (1-\varepsilon)/15$ and $t_B = (1-\varepsilon)/10$. After multiplying each balance equation by its denominator $d_j - 1 + \mu$, the equations become

$$\begin{aligned} (r-\varepsilon\mu)a - \varepsilon c - \varepsilon f &= rt_A, & -\varepsilon a + rb - \varepsilon\mu c - \varepsilon f &= rt_A, \\ -\varepsilon\mu b + wc - \varepsilon d &= wt_B, & -\varepsilon b + (w-\varepsilon\mu)d &= wt_B, \\ -\varepsilon a - \varepsilon c + rh - \varepsilon\mu f &= rt_A, & -\varepsilon h + f &= t_B. \end{aligned} \tag{7}$$

In the last equation the two incoming masses at the roof are equal, so the ordinary and return weights combine to w. The roof score is $2f$. Let M be the coefficient matrix in the order (a, b, c, d, h, f) and $b_{30} = 30(rt_A, rt_A, wt_B, wt_B, rt_A, t_B)^T$. Expansion gives the polynomial identities

$$\det M = (\mu+1)(\mu+2)(1-\varepsilon)D_\varepsilon(s), \qquad \det M[f \leftarrow b_{30}] = (\mu+1)(\mu+2)(1-\varepsilon)N_\varepsilon(s),$$

which the code checks coefficient by coefficient over $\mathbb{Q}[\mu, \varepsilon]$. Cramer's rule gives $f = N_\varepsilon/(30D_\varepsilon)$. □

4 The complete damping classification of the roof vertex

Theorem 4.1 (One threshold, one minimum)**.** *Let ε_c be the unique root in $(0, 1/4)$ of*

$$A(\varepsilon) = \varepsilon^6 - 2\varepsilon^5 - 9\varepsilon^4 - 11\varepsilon^3 - 16\varepsilon^2 - 12\varepsilon + 4, \tag{8}$$

so that $2406813 \cdot 10^{-7} < \varepsilon_c < 2406814 \cdot 10^{-7}$. For the house roof:

1. *if $0 < \varepsilon \le \varepsilon_c$, then $\pi_\mu(4)$ is strictly increasing on $[0, \infty)$;*
2. *if $\varepsilon_c < \varepsilon < 1$, then $\pi_\mu(4)$ decreases to exactly one strict minimum at some $\mu_*(\varepsilon) > 0$ and then increases;*
3. *for $\varepsilon > \varepsilon_c$, the minimum lies below, at or above $\mu = 1$ according as $\varepsilon < 4/9$, $\varepsilon = 4/9$ or $\varepsilon > 4/9$. Since $\varepsilon_c < 4/9$, each case occurs.*

Proof. Let $B(\varepsilon) = 4\varepsilon^5 - 4\varepsilon^4 - 14\varepsilon^3 - 18\varepsilon^2 - 16\varepsilon + 12$, $C(\varepsilon) = 6\varepsilon^4 - \varepsilon + 13$ and

$$R_\varepsilon(s) = (1+\varepsilon)^2 s^4 + 2(1+\varepsilon)(2\varepsilon^2+3)s^3 + C(\varepsilon)s^2 + B(\varepsilon)s + A(\varepsilon).$$

Differentiating (6) gives the polynomial identity

$$\frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(4) = \frac{\varepsilon^2(1-\varepsilon)R_\varepsilon(s)}{15D_\varepsilon(s)^2}. \tag{9}$$

The three leading coefficients of R_ε are positive, since $C(\varepsilon) > 12$. On $(0, 1)$ we have $A'(\varepsilon) = 6\varepsilon^5 - 10\varepsilon^4 - 36\varepsilon^3 - 33\varepsilon^2 - 32\varepsilon - 12 < 0$, because $6\varepsilon^5 \le 6\varepsilon^4 < 10\varepsilon^4$. Also $A(0) = 4$ and $A(1/4) < 0$, so A has one root ε_c in $(0, 1/4)$ and is negative on $(\varepsilon_c, 1)$. Exact substitution gives the bracket. For $0 < \varepsilon \le 1/4$, $B(\varepsilon) \ge 12 - 16/4 - 18/16 - 14/64 - 4/256 > 0$. Hence, if $\varepsilon \le \varepsilon_c$, all coefficients of R_ε are nonnegative and only the constant one can vanish, and $\pi_\mu(4)$ is strictly increasing. If $\varepsilon > \varepsilon_c$, the constant coefficient is negative and the three leading ones are positive, so the coefficient sequence has exactly one sign change whatever the sign of B. By Descartes' rule there is exactly one positive root, and it is simple. The derivative changes from negative to positive there. Finally, $[(\partial_\mu N)D - N(\partial_\mu D)]_{\mu=1} = 9\varepsilon^2(1-\varepsilon)(4-9\varepsilon)$ with $N = N_\varepsilon((1-\varepsilon)\mu)$ and $D = D_\varepsilon((1-\varepsilon)\mu)$, which locates the minimum relative to $\mu = 1$. □

Corollary 4.2 (Failure inside the original range)**.** *Monotonicity fails even for μ restricted to $[0, 1]$: at $\varepsilon = 1/3$, $\pi_0(4) = 3929/21570$, $\pi_{1/2}(4) = 1481/8133$ and $\pi_1(4) = 51/280$, and the middle value is strictly the smallest.*

Proposition 4.3 (Exact failure range of the endpoint shortcut)**.** *Let ε_0 be the unique root in $(0,1)$ of $F(\varepsilon) = 2\varepsilon^4 + \varepsilon^3 + 6\varepsilon^2 + 10\varepsilon - 4$, so that $3291037 \cdot 10^{-7} < \varepsilon_0 < 3291038 \cdot 10^{-7}$. For every ε, $\pi_\infty(4) > \pi_1(4)$. Moreover $\pi_0(4) > \pi_1(4)$ precisely when $\varepsilon > \varepsilon_0$.*

Proof. One has $\pi_1(4) = (6-\varepsilon)/[5(6+\varepsilon-\varepsilon^2)]$ and $\pi_\infty(4) = (3+2\varepsilon)/[15(1+\varepsilon)]$, so

$$\pi_\infty(4) - \pi_1(4) = \frac{2\varepsilon^2(1-\varepsilon)}{15(1+\varepsilon)(6+\varepsilon-\varepsilon^2)} > 0, \qquad \pi_0(4) - \pi_1(4) = \frac{\varepsilon^2(1-\varepsilon)F(\varepsilon)}{15d_0(6+\varepsilon-\varepsilon^2)}.$$

Since $F(0) < 0 < F(1)$ and $F' > 0$, the zero is unique. □

Remark 4.4 (Two different thresholds). For $\varepsilon_c < \varepsilon \le \varepsilon_0$ the roof is nonmonotone, yet the shortcut still gives the right side for π_0, because the minimum is shallow at $\mu = 0$. The directional error needs $\varepsilon > \varepsilon_0$. It includes the conventional damping $\varepsilon = 17/20$, where $\pi_0(4) = 12276466/72640495$, $\pi_1(4) = 412/2451$ and $\pi_\infty(4) = 94/555$.

5 Smallest counterexamples and large examples

Two further graphs appear (Figure 2). The *gem* is the path 0–1–2–3 together with an apex 4 adjacent to all four path vertices. The *paw* is a triangle $1, 2, 3$ with a pendant vertex 0 attached at 3.

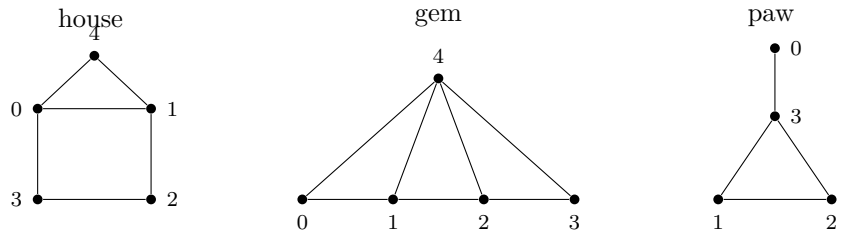


Figure 2: The house, the gem and the paw, labelled as in the text and the code.

Proposition 5.1 (The gem fails at small damping)**.** *For the gem with uniform restart, write $\frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(1) = \varepsilon^2(\varepsilon-1)R(\mu,\varepsilon)/S(\mu,\varepsilon)$ with $S > 0$ the square of the score denominator. Then R has degree six in μ with leading coefficient $4(1-\varepsilon^2)^4$,*

$$R(\mu, 0) = (\mu+2)^2(4\mu^4 + 22\mu^3 + 37\mu^2 + 12\mu - 11),$$

$$R(0, \varepsilon) = 4\varepsilon^8 + 8\varepsilon^7 - 16\varepsilon^6 - 40\varepsilon^5 - 15\varepsilon^4 + 63\varepsilon^3 + 148\varepsilon^2 + 64\varepsilon - 44.$$

The polynomial $R(0,\cdot)$ is negative on $(0, \varepsilon_g)$, where $\varepsilon_g \in (0.3565987, 0.3565988)$ is its first positive root. Consequently, for every $0 < \varepsilon < \varepsilon_g$, the score $\pi_\mu(1) = \pi_\mu(2)$ increases at $\mu = 0$ and decreases for all large μ, so it is nonmonotone.

Proof. The scores are obtained by exact LU solution of the 14-state system over $\mathbb{Q}(\mu,\varepsilon)$. The displayed polynomials, the positivity of S (a square times a positive constant), and the absence of roots of $R(0,\cdot)$ in $(0, 0.3565987)$ are certified by exact Sturm root counts (Section 12). If $R(0,\varepsilon) < 0$, the derivative is positive at $\mu = 0$. The leading coefficient is positive for $\varepsilon < 1$, so $R(\mu,\varepsilon) > 0$ and the derivative is negative for all large μ. Exact witnesses: at $\varepsilon = 1/10$, the values $\pi_\mu(1)$ at $\mu = 0, 1/3, 1$ are $53729499/266457580$, $4997753/24784610$ and $123/610$, and the middle one is strictly the largest. At $\varepsilon = 1/4$, the values at $\mu = 0, 1/10, 1$ are $263957/1293580$, $263643/1292020$ and $51/250$. □

Theorem 5.2 (Smallest order, for every damping)**.** *For every fixed $\varepsilon \in (0,1)$, five is the smallest order of a connected simple graph with minimum degree at least two on which some coordinate of π_μ (uniform restart) is nonmonotone in $\mu \in [0,\infty)$. A witness is the gem when $\varepsilon \le 1/3$ and the house when $\varepsilon > \varepsilon_c$.*

Proof. Since $\varepsilon_c < 1/4 < 1/3 < \varepsilon_g$, Proposition 5.1 and Theorem 4.1 cover every ε. At order three the only eligible graph is K_3. At order four the complement has maximum degree one, so the eligible graphs are C_4, K_4 and the diamond $K_4 - e$. Regular graphs have μ-independent scores (Theorem 9.1). In the diamond (hubs $0,1$, tips $2,3$), let a, b, c be the hub-to-hub, hub-to-tip and tip-to-hub masses. With $r = \mu + 2$, the orbit balances

$$(r-\varepsilon\mu)a - 2\varepsilon c = \tfrac{r(1-\varepsilon)}{12}, \quad -\varepsilon a + rb - \varepsilon(\mu+1)c = \tfrac{r(1-\varepsilon)}{12}, \quad -\varepsilon b + c = \tfrac{1-\varepsilon}{8}$$

give the tip score

$$z_\mu = 2c = \frac{(3-\varepsilon-2\varepsilon^2)\mu + 2\varepsilon^2 + 4\varepsilon + 6}{12[(1-\varepsilon^2)\mu + \varepsilon^2 + 2\varepsilon + 2]}, \qquad z'_\mu = \frac{\varepsilon^2(1-\varepsilon)}{12[(1-\varepsilon^2)\mu + \varepsilon^2 + 2\varepsilon + 2]^2} > 0,$$

and the hub scores equal $1/2 - z_\mu$. Every coordinate is therefore strictly monotone. □

Proposition 5.3 (Leaves allowed: the paw is a smallest counterexample)**.** *With leaves allowed (Section 2), every coordinate is independent of μ on K_2, K_3 and the path P_3 (for P_3, $\pi = ((2+\varepsilon), 2(1+2\varepsilon), (2+\varepsilon))/(6(1+\varepsilon))$). On the paw with uniform restart, every coordinate has a derivative in μ whose numerator is a quadratic $a_2\mu^2 + a_1\mu + a_0$ (times $\varepsilon^2(1-\varepsilon)$) with a_2, a_1 of one fixed sign on $(0,1)$. Let $\varepsilon_1 \in (0.0783254, 0.0783255)$ be the root in $(0,1)$ of $2\varepsilon^4+10\varepsilon^3+9\varepsilon^2+12\varepsilon-1$, and $\varepsilon_2 \in (0.4355202, 0.4355203)$ the root in $(0,1)$ of $\varepsilon^4 - 6\varepsilon^3 + 2\varepsilon^2 - 9\varepsilon + 4$. Then:*

1. *the leaf 0 is monotone for every ε;*
2. *the degree-two vertices $1, 2$ are nonmonotone exactly when $\varepsilon < \varepsilon_1$;*
3. *the hub 3 is nonmonotone exactly when $\varepsilon > \varepsilon_2$.*

Thus, without the degree restriction, order four is the smallest order with a nonmonotone coordinate for $\varepsilon \in (0,\varepsilon_1) \cup (\varepsilon_2, 1)$ (Corollary 5.4). At $\varepsilon = 1/2$ the hub values at $\mu = 1/10, 1/2, 1$ are $2148/6389$, $344/1023$ and $39/116$, an interior maximum inside $(0,1]$.

Proof. P_3 is bipartite with parts of constant degree, and the displayed scores follow from the exact solve. For the paw, the coefficients are listed in the results file. In each case a_2 and a_1 share a constant sign on $(0,1)$, certified by exact Sturm root counts. The quadratic then has a positive root, necessarily simple, if and only if a_0 has the opposite sign, by Descartes' rule. The leaf's a_0 has the sign of a_2. For vertices 1 and 3, a_0 changes sign exactly once in $(0,1)$, at ε_1 and ε_2 respectively, with the stated orientation. □

Corollary 5.4 (Smallest order with leaves allowed)**.** *Let $n_{\min}(\varepsilon)$ be the smallest order of a connected simple graph with a coordinate of π_μ (uniform restart) nonmonotone in $\mu \in (0,\infty)$. Then $n_{\min}(\varepsilon) = 4$ for $0 < \varepsilon < \varepsilon_1$ and for $\varepsilon_2 < \varepsilon < 1$, and $n_{\min}(\varepsilon) = 5$ for $\varepsilon_1 \le \varepsilon \le \varepsilon_2$. With minimum degree at least two the smallest order is five for every ε (Theorem 5.2).*

Proof. Orders two and three are μ-independent by Proposition 5.3. The connected graphs of order four are the star $K_{1,3}$, C_4, K_4, the diamond, the path P_4 and the paw. The star is bipartite biregular and C_4, K_4 are regular, so their scores do not depend on μ (exact symbolic solves; for $K_{1,3}$ the argument of Theorem 9.1(iv)⇒(i) applies verbatim, since a leaf returns with probability one). The diamond is monotone by the proof of Theorem 5.2. For P_4 (path 0–1–2–3) the exact solve over $\mathbb{Q}(\mu,\varepsilon)$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(0) = \frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(3) = -\frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(1) = -\frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(2) = \frac{\varepsilon^2(1-\varepsilon)}{8[(\varepsilon^2-1)\mu - \varepsilon^2 - \varepsilon - 1]^2} > 0,$$

so every coordinate is strictly monotone. Only the paw remains, and Proposition 5.3 gives its exact failure set. Order five suffices for every ε by Theorem 5.2. □

The house and the gem cover overlapping damping ranges; no single five-vertex graph is claimed to fail at every ε.

Proposition 5.5 (A tree with a nonmonotone leaf)**.** *Take the tree with edges* $04, 13, 23, 34$*, uniform restart and* $\varepsilon = 1/2$*. Its leaf* 1 *has*

$$\pi_\mu(1) = \frac{21\mu^2 + 95\mu + 84}{15(9\mu^2 + 42\mu + 37)}, \qquad \frac{\mathrm{d}}{\mathrm{d}\mu}\pi_\mu(1) = \frac{27\mu^2 + 42\mu - 13}{15(9\mu^2 + 42\mu + 37)^2},$$

so $\pi_\mu(1)$ *decreases to a strict minimum at* $\mu = (2\sqrt{22} - 7)/9 \approx 0.2645$ *and then increases. An exact certificate inside* $(0, 1)$*:* $\pi_{1/10}(1) = 9371/61935$*,* $\pi_{1/4}(1) = 349/2307$ *and* $\pi_{1/2}(1) = 547/3615$*, with successive differences* $-398/15876005$ *and* $98/2779935$*.*

Proof. The rational response is the interpolated one of Section 7; the code checks the displayed closed form as a polynomial identity against it, and the three values by independent full eight-state solves. No convention at $\mu = 0$ is involved. □

Lemma 5.6 (Lifting through a finite cover)**.** *Let* $\widetilde{G}$ *be a* k*-sheeted cover of* G*, and lift the restart by assigning* v_i/k *to each vertex over* i*. Then the lifted edge and vertex masses are* $q_\mu(i, j)/k$ *and* $\pi_\mu(i)/k$ *on every lift.*

Proof. A covering map is a bijection on each neighbourhood. It therefore preserves degrees, the unique return, and the weights (2). The proposed masses satisfy every lifted balance equation and sum to one; uniqueness concludes. This is an instance of PageRank invariance under graph fibrations [8] (lumpability [18]) applied to the edge chain. □

Corollary 5.7 (A positive fraction of failing vertices)**.** *For every* $k \geq 2$ *there is a connected simple graph on* $5k$ *vertices, with degrees two and three, in which* k *vertices have* $1/k$ *times the house roof response. Their normalised scores* $5k\,\pi_\mu$ *equal* $5\pi_\mu(4)$*. In particular one fifth of the vertices are nonmonotone whenever* $\varepsilon > \varepsilon_c$*. Covers of the gem give graphs on* $5k$ *vertices, with degrees two, three and four, in which at least two fifths of the vertices (the lifts of gem vertices* 1 *and* 2*) are nonmonotone for* $\varepsilon < \varepsilon_g$*.*

Proof. Take vertices (i, t), $t \in \mathbb{Z}/k$. Lift every house edge within its sheet, except that 01 becomes $(0, t)(1, t+1)$. Removing 01 leaves the house connected, so each sheet is connected, and the shifted edges join consecutive sheets. Apply Lemma 5.6. For the gem, shift the edge 01 in the same way; removing it leaves the gem connected. □

6 Why vertices dip, and cheap tests

6.1 A two-step picture

Expand $q_\mu = (1 - \varepsilon)\sum_t \varepsilon^t Q_\mu^t u$. The tail of a state after t steps is distributed as $p_t(\cdot; \mu)$, where $p_0 = v$ and $p_1 = Pv$. Neither depends on μ, because the first move out of the restart edge is uniform. The first μ-dependence appears at $t = 2$: the walk has moved $i \to j$ and now chooses between returning to i and moving on. Under uniform restart, the mass arriving at k after two steps is $p_2(k; \mu) = f_k(\mu)/n$, where

$$f_k(\mu) = \sum_{j \sim k} \frac{d_k h_j - 1 + \mu}{d_k(d_j - 1 + \mu)}, \qquad h_j = \sum_{l \sim j} \frac{1}{d_l}. \tag{10}$$

Each term is a Möbius function of μ, increasing or decreasing according to the sign of

$$\sigma_j = \frac{d_j}{d_k} - h_j.$$

So raising μ sends more two-step mass back to k from a neighbour j when j's other neighbours have high degree (small h_j), and less when they have low degree. When the neighbours of k disagree in sign, and their poles $d_j - 1$ differ, the two-step profile can rise and then fall. That is the gem mechanism.

Theorem 6.1 (Small-damping criterion)**.** *Assume uniform restart and minimum degree at least two. Suppose* $0 \le a < b < c$ *and* $\delta = f_k(b) - \max\{f_k(a), f_k(c)\} > 0$ *(respectively* $\delta = \min\{f_k(a), f_k(c)\} - f_k(b) > 0$*). Then* $\pi_\mu(k)$ *is nonmonotone in* μ *for every* $0 < \varepsilon < \delta/(n+\delta)$*. If all* σ_j*,* $j \sim k$*, have the same weak sign, then* f_k *is monotone, and any nonmonotonicity of* $\pi_\mu(k)$ *is a phenomenon of order* ε^3 *or higher.*

Proof. We have $\pi_\mu(k) = (1-\varepsilon)[v_k + \varepsilon(Pv)_k + \varepsilon^2 f_k(\mu)/n] + \rho_\mu$ with $0 \le \rho_\mu = (1-\varepsilon)\sum_{t\ge 3} \varepsilon^t p_t(k;\mu) \le \varepsilon^3$. Hence $\pi_b(k) - \pi_a(k) \ge (1-\varepsilon)\varepsilon^2\delta/n - \varepsilon^3 > 0$ when $(1-\varepsilon)\delta > n\varepsilon$, that is, $\varepsilon < \delta/(n+\delta)$, and similarly for c. The function $(\alpha+\mu)/(\gamma+\mu)$ has derivative of the sign of $\gamma - \alpha$; here $\gamma - \alpha = d_j - d_k h_j = d_k \sigma_j$. □

The roof. Both neighbours of the house roof have $\sigma = 3/2 - 4/3 = 1/6 > 0$, so its two-step profile is increasing. The roof's dip is a third-order effect, which is why it needs damping above ε_c. Heuristically, only longer walks see the competition between the shortcut 0–1 (avoided when returns are suppressed) and the path 1–2–3–0 (whose degree-two vertices funnel non-backtracking walkers forward). For gem vertex 1 the three neighbours give $\sigma = 1/12, -1/12, -1/3$, and the dip already appears at second order. In the census (Section 7), 276 of the 3985 vertices of minimum-degree-two graphs have a nonmonotone two-step profile; 226 of these are nonmonotone at $\varepsilon = 1/10$, out of 256 such vertices in all. No vertex with uniform σ signs has a nonmonotone profile, as the theorem predicts.

6.2 The slope at ordinary PageRank

Proposition 6.2 (Exact slope at $\mu = 1$)**.** *Let the minimum degree be at least two, and let* v *be any restart vector. With* $h = P\mathbf{1}$ *and* $\circ$ *the entrywise product,*

$$\frac{\mathrm{d}\pi_\mu}{\mathrm{d}\mu}\Big|_{\mu=1} = \varepsilon^2 (I - \varepsilon P)^{-1} [h \circ D_G^{-1}\pi_1 - P D_G^{-1} P \pi_1].$$

Proof. Differentiating (3) gives $\partial_\mu q = \varepsilon(I - \varepsilon Q_\mu)^{-1} Q'_\mu q_\mu$. At $\mu = 1$, $q_1(i,j) = \pi_1(i)/d_i$ and $\partial_\mu (Q_\mu)_{(j,k),(i,j)} = \mathbf{1}_{\{k=i\}}/d_j - 1/d_j^2$. Hence $g = Q'_1 q_1$ satisfies $g(j,k) = \pi_1(k)/(d_k d_j) - (P\pi_1)_j/d_j^2$. Its tail sums vanish, and its head sums form the vector $G = h \circ D_G^{-1}\pi_1 - P D_G^{-1} P\pi_1$. Let $y = (I - \varepsilon Q_1)^{-1} g$. Since $(Q_1 x)(j,k)$ equals the head sum of x at j divided by d_j, the head sums Y of y satisfy $Y = G + \varepsilon P Y$, and the tail sums of y equal εY. Therefore $\partial_\mu \pi = \varepsilon \cdot \varepsilon Y = \varepsilon^2 (I - \varepsilon P)^{-1} G$. □

Corollary 6.3 (Three endpoint tests)**.** *Each of the following implies that* $\pi_\mu(i)$ *is nonmonotone, with* $\pi'_1 = \partial_\mu \pi_\mu|_{\mu=1}$*: (T1)* $(\pi_0(i) - \pi_1(i))(\pi_\infty(i) - \pi_1(i)) > 0$*; (T2)* $\pi'_1(i)\,(\pi_\infty(i) - \pi_1(i)) < 0$*; (T3)* $\pi'_1(i)\,(\pi_1(i) - \pi_0(i)) < 0$*, in which case the failure occurs on* $[0,1]$*.*

Tests (T2) and (T3) cost two $n \times n$ solves plus π_0 or π_∞. Over the census graphs of minimum degree two at $\varepsilon = 1/2$, the formula of Proposition 6.2 agrees exactly with the derivative of the certified rational responses at all 3985 vertices. The tests detect 285 of the 478 nonmonotone vertices, and they never fire on a monotone vertex, as they cannot.

7 A certified census on at most seven vertices

Method. For each connected graph on 2 to 7 vertices (the networkx graph atlas lists all graphs up to isomorphism on at most seven vertices), and each $\varepsilon \in \mathcal{D} = \{1/10, 1/4, 1/3, 1/2, 17/20\}$, we compute every score exactly as a rational function $\pi_\mu(i) = N_i(\mu)/\mathrm{Den}(\mu)$. Here Den is the

determinant of the $2m \times 2m$ edge system with each row scaled by $d_j - 1 + \mu$ (leaf rows are unscaled). Its entries have degree at most one in μ, so $\deg N_i, \deg \mathrm{Den} \le 2m$, and $\mathrm{Den} \not\equiv 0$ on $[0, \infty)$. We evaluate exactly at $\mu = 0, \ldots, 2m$ using a block elimination to an $n \times n$ system, interpolate, and confirm at two further points. A coordinate is nonmonotone on an open interval if and only if $W_i = N_i' \mathrm{Den} - N_i \mathrm{Den}'$ has a root of odd multiplicity there. We take the squarefree decomposition of W_i and form the product O_i of its odd-multiplicity factors. Any root of O_i at $\mu = 0$ or $\mu = 1$ is divided out exactly, and the roots of the quotient in the open intervals $(0, \infty)$ and $(0, 1)$ are counted by an exact Sturm sequence. A root at $\mu = 1$ counts for $(0, \infty)$ but not for $(0, 1)$, and a root at $\mu = 0$ counts for neither. The counter is checked on known-root regressions, including all 128 subsets of $\{-2, -\frac{1}{3}, 0, \frac{1}{2}, 1, 2, \frac{5}{2}\}$. Common factors of N_i and Den have no roots in $[0, \infty)$ and enter W_i squared, so they do not affect the decision. No floating-point arithmetic is used.

Table 3: Certified census (uniform restart): number of connected graphs with at least one nonmonotone coordinate on $\mu \in [0, \infty)$; in parentheses, those with a coordinate nonmonotone on $[0, 1]$. "Some" and "all" refer to the five damping values. For graphs with leaves the range is $(0, \infty)$, respectively $(0, 1]$.

class	n	graphs	$\varepsilon = \frac{1}{10}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{17}{20}$	some / all
min. deg. ≥ 2	3	1	0	0	0	0	0	0 / 0
	4	3	0	0	0	0	0	0 / 0
	5	11	1 (1)	2 (2)	2 (2)	1 (0)	1 (0)	2 / 0
	6	61	13 (2)	17 (6)	16 (5)	17 (5)	23 (1)	27 / 9
	7	507	150 (50)	186 (68)	207 (75)	261 (85)	321 (44)	353 / 107
all connected	2–3	3	0	0	0	0	0	0 / 0
	4	6	0	0	0	1 (1)	1 (0)	1 / 0
	5	21	3 (3)	5 (4)	5 (2)	6 (3)	8 (0)	9 / 0
	6	112	29 (9)	38 (15)	44 (17)	48 (12)	62 (7)	68 / 19
	7	853	282 (113)	392 (141)	444 (156)	543 (172)	637 (89)	675 / 214

Findings. At order five with minimum degree two, the nonmonotone graphs at the five damping values are exactly the house and the gem: the gem at $\varepsilon \le 1/3$, the house at $\varepsilon \ge 1/4$. Which of the two appears therefore depends on the damping: at $\varepsilon \in \{1/2, 17/20\}$ only the house fails, and at small damping the gem fails as well. Proposition 5.1 and Theorem 5.2 give the picture for every ε. At order four the only failure is the paw, at $\varepsilon = 1/2$ and $17/20$, in agreement with Proposition 5.3. At order seven, 70% of minimum-degree-two graphs (353 of 507) have a nonmonotone coordinate for some damping in $\mathcal{D}$. At $\varepsilon = 17/20$, 632 of their 3549 vertices (18%) are nonmonotone. Monotonicity is therefore atypical among seven-vertex graphs at these damping values. The census is certified for the five listed damping values only; the dependence on ε in between is not certified.

8 The spread clause

Proposition 8.1 (Both readings of the spread clause fail)**.** *1. As printed (spread nondecreasing in μ), the clause fails on the diamond for every ε: the spread equals $\frac{1}{2} - 2z_\mu$, which is positive and strictly decreasing. It also fails on the house: at $\varepsilon = 1/2$ the spread takes the values* 13/241, 24004/446965, 836/15815, 13/250, 301/5935, 59/1214, 5165/112301 *and* 2/45 *at $\mu = 0, 1/10, 1/2, 1, 2, 5, 20, \infty$, a strictly decreasing sequence.*

2. The reversed reading (spread nonincreasing) also fails. On the six-vertex graph with edges $01, 04, 05, 12, 23, 34, 35$ *(minimum degree two) at $\varepsilon = 17/20$, the spread is* 702355/10672572 *at $\mu = 0$ and* 24034345/365205604 *at $\mu = 1/10$, which is strictly larger.*

Proof. For the diamond, the hub score $\frac{1}{2} - z_\mu$ exceeds the tip score z_μ, because $z_\mu < 1/4$: this reduces to $(\varepsilon - \varepsilon^2)\mu + \varepsilon^2 + 2\varepsilon > 0$. Since $z'_\mu > 0$, the spread decreases. The remaining values are exact rational solves. □

The printed inequality is probably a reversal of the intended statement, since the source's own heuristic has high-degree scores falling and low-degree scores rising. A grid search over the 61 six-vertex minimum-degree-two graphs at $\varepsilon \in \{1/10, 1/2, 17/20\}$ (17 values of μ; exact arithmetic at the grid points) found reversed-clause failures in two graphs, both at $\varepsilon = 17/20$. This is a discovery search, not a certificate: failures between grid points are not excluded. Only the displayed witness is certified. Under either reading the conjunction in Conjecture 1 is false. Independently of the spread clause, the house refutes the coordinatewise clause.

A natural repair of the reversed clause would require the maximising and minimising vertices to be unique. That is not enough.

Proposition 8.2 (Unique extrema do not force decreasing spread)**.** *Take the seven-vertex graph with edges* $03, 04, 12, 13, 14, 25, 26, 34, 56$ *(minimum degree two), uniform restart and* $\varepsilon = 17/20$. *For every* $\mu \in [0, \infty)$ *and at* $\mu = \infty$, *vertex* 2 *is the unique maximiser and vertex* 0 *the unique minimiser of* π_μ. *Nevertheless the spread increases:*

$$\pi_0(2) - \pi_0(0) = \frac{2874404080966}{51723978657253} < \frac{3144178174777409 54}{5620343572784081337} = \pi_{1/10}(2) - \pi_{1/10}(0).$$

Proof. For each of the twelve gaps $\pi_\mu(2) - \pi_\mu(j)$, $j \neq 2$, and $\pi_\mu(j) - \pi_\mu(0)$, $j \neq 0$, the code reduces the exact rational response to lowest terms. Every reduced numerator and denominator has strictly positive coefficients, so every gap is positive for all $\mu \geq 0$; the gaps at $\mu = \infty$ are positive by direct evaluation of (4). The two spreads are independent full-state solves; their difference is $154002826624500199512299582 60/4152950442930184229546491385 5323 > 0$. □

9 Exact invariance

Two vertices lie in the same *period class* when an even-length walk joins them. A connected non-bipartite graph has one class; a bipartite graph has its two sides.

Theorem 9.1 (Complete backtracking invariance)**.** *For a connected graph of minimum degree at least two, a probability vector* v *and a fixed* $0 < \varepsilon < 1$, *the following are equivalent: (i)* π_μ *is independent of* $\mu \in [0, \infty)$*; (ii)* $\pi_1 = \pi_\infty$*; (iii)* $P^2 v = v$*; (iv)* v_i/d_i *is constant on each period class. Under uniform restart these hold exactly for regular and bipartite biregular graphs.*

Proof. (i)⇒(ii) by taking the limit. Multiplying (ii) by the invertible matrix $(1 + \varepsilon)(I - \varepsilon P)$ gives $(1 - \varepsilon^2)v = v - \varepsilon^2 P^2 v$, which is (iii). With $w = D_G^{-1} v$ and $R = D_G^{-1} A_G$, (iii) is $R^2 w = w$. On each period class R^2 is an irreducible stochastic matrix (bipartiteness is the only obstruction to irreducibility of R^2), and its fixed vectors are constant, which is (iv). For (iv)⇒(i), put $q(i, j) = (w_i + \varepsilon w_j)/(1 + \varepsilon)$. All neighbours of a vertex lie in one class, so $(Q_\mu q)(i, j) = q(j, i)$ for every μ, and $q(i, j) = (1 - \varepsilon)w_i + \varepsilon q(j, i)$. Hence q solves (3) for all μ. □

Sufficiency for uniform restart on regular graphs goes back to [5, Theorem 4] at $\mu = 0$ (as noted in [2]) and to [2, Theorems 3.1–3.2] for $\mu > 0$. The converse, the arbitrary-restart form, and the inclusion of $\mu = 0$ appear to be new, though elementary. Admissible restarts are $v_i = d_i/(2m)$ on non-bipartite graphs, and $v_i = td_i/m$ on V_A, $(1 - t)d_i/m$ on V_B $(0 \leq t \leq 1)$ on bipartite graphs with m edges. The path P_3 of Proposition 5.3 is a bipartite biregular example with leaves.

Remark 9.2 (Residual identity, and what it does not give). Subtracting the formulas (4) gives $\pi_1 - \pi_\infty = \frac{\varepsilon^2}{1+\varepsilon}(I - \varepsilon P)^{-1}(P^2 v - v)$. Hence

$$\frac{\varepsilon^2}{(1+\varepsilon)^2}\|P^2 v - v\|_1 \leq \|\pi_1 - \pi_\infty\|_1 \leq \frac{\varepsilon^2}{1 - \varepsilon^2}\|P^2 v - v\|_1.$$

The upper bound is not a practical certificate for pairwise order. It is of order one while score gaps are of order $1/n$, and on every network of Section 10 it certified no pair (numerical). A coordinatewise version, $|\pi_1 - \pi_\infty| \le \frac{\varepsilon^2}{1+\varepsilon}(I - \varepsilon P)^{-1}|P^2 v - v|$, costs one more solve of the same size as π_1. The identity is useful mainly to show that the gap is of order ε^2, and to measure how far a graph is from the invariant case. It says nothing about the non-backtracking endpoint.

10 Prevalence and practical impact (numerical)

All numbers in this section are floating-point, produced by `code/prevalence.py`. Each graph is replaced by the largest component of its 2-core, the restart is uniform, and μ runs over 300 grid points in $[0, 1000]$ plus $\mu = \infty$. A vertex counts as nonmonotone when consecutive grid differences of both signs exceed $10^{-12} \max \pi$. The *excursion* is the overshoot beyond the endpoint range, relative to $\pi_1(i)$. "Shortcut wrong" counts vertices whose π_0 and π_∞ lie on the same strict side of π_1. Kendall's τ compares rankings.

Table 4: Prevalence of nonmonotone vertices (numerical; 2-cores; uniform restart). Fractions of vertices. Excursion: median / maximum over nonmonotone vertices, relative to $\pi_1(i)$. Reversals: pairs whose order at some grid point differs from the common order at $\mu = 0$ and $\mu = \infty$. ER: G(200, 0.02); BA: Barabási–Albert ($m = 2$); HK: Holme–Kim; CM: configuration model with power-law degrees.

graph	n	ε	nonmon.	on $[0,1]$	excursion	shortcut wrong	$\tau(\pi_0, \pi_1)$	reversals
Karate	33	0.85	0.36	0.12	0.003 / 0.020	0.21	0.99	0
Les Misérables	59	0.50	0.10	0.02	0.002 / 0.004	0.05	0.96	4
		0.85	0.22	0.12	0.001 / 0.055	0.14	0.97	1
ER	181	0.85	0.29	0.04	0.005 / 0.045	0.13	0.99	23
BA	300	0.85	0.41	0.09	0.013 / 0.104	0.25	0.97	122
HK	300	0.85	0.39	0.08	0.011 / 0.104	0.26	0.97	126
CM	299	0.85	0.41	0.08	0.015 / 0.119	0.25	0.95	310
3-regular	100	any	0	0	–	0	–	0

Over 50 random graphs $G(n, 4/n)$ for each of $n = 20$ and $n = 100$, the setting of the source's simulations, the mean fraction of nonmonotone vertices was 0.16 and 0.18 at $\varepsilon = 1/2$, and 0.25 and 0.30 at $\varepsilon = 0.85$. Every graph with $n = 100$ had a nonmonotone vertex. The picture is consistent. Apart from the ten-vertex Florentine families graph (no failure at $\varepsilon = 1/2$), nonmonotone vertices are common (10–41%), mostly with the turning point at $\mu > 1$, and the endpoint shortcut gives the wrong side for up to 26% of vertices. The excursions are small, with medians between 0.004% and 1.5% of the score, which plausibly explains why plots on a 20-point grid looked monotone. Rankings are stable ($\tau \ge 0.93$, identical top-10 sets except one), so the failure threatens score-side inferences far more than rank-based ones. The cheap proxy π_∞ is a poor stand-in for π_1 in ranking ($\tau(\pi_1, \pi_\infty)$ between 0.74 and 0.96). Nonmonotone vertices had the same median degree as their graphs, except in Karate (4 against 3), so degree alone does not predict them. At small damping, the two-step signs σ_j of Section 6 are a local indicator; we have not tested them on these networks.

Proposition 10.1 (An exact transient rank reversal)**.** *In the largest component of the 2-core of the unweighted Les Misérables co-occurrence graph of Knuth's Stanford GraphBase [20], as distributed with networkx 3.6.1 [15] (77 vertices and 254 edges; core $n = 59$, $m = 236$) at $\varepsilon = 1/2$, the vertices Fauchelevent (F) and Pontmercy (P) satisfy $\pi_0(F) < \pi_0(P)$, $\pi_\infty(F) < \pi_\infty(P)$ (difference $-4238/26584515$), but $\pi_1(F) > \pi_1(P)$.*

The proof is an exact rational solve; the fractions are in the results file. Thus ordinary PageRank can rank a pair differently from both backtracking extremes. By contrast, a finite search in exact arithmetic over the 507 seven-vertex graphs of minimum degree two, at $\varepsilon \in \{\frac{1}{2}, \frac{17}{20}\}$

and $\mu \in \{\frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, 16\}$, found no pair ordered strictly the same way at $\mu = 0$ and $\mu = \infty$ and strictly the other way at a grid point (124208 pair–grid comparisons). This covers the grid points only; it does not exclude reversals between them.

11 Degree tails do not determine infinite-backtracking tails

A finite graph has bounded degrees, so Conjecture 2 must concern a graph sequence and the score of a uniformly chosen vertex, normalised as $n\pi_\infty$ (mean one); this is the local-weak-convergence reading of [11, 17]. It is one reasonable formalisation. The counterexample below has a bounded score law, so it refutes any reading that requires unbounded scores. Under uniform restart, (4) gives

$$n\pi_\infty(i) = \frac{1 + \varepsilon h_i}{1 + \varepsilon}, \qquad h_i = \sum_{j \sim i} \frac{1}{d_j}. \tag{11}$$

Proposition 11.1 (A one-sided bound)**.** *If the minimum degree is at least two, then $h_i \le d_i/2$ and $n\pi_\infty(i) \le (1 + \varepsilon d_i/2)/(1 + \varepsilon)$. Hence, along any graph sequence, the tail of $n\pi_\infty(U)$ is never heavier than that of the degree: $\mathbb{P}(n\pi_\infty(U) > x) \le \mathbb{P}(d_U > 2((1 + \varepsilon)x - 1)/\varepsilon)$.*

Large degree alone does not force a large score, because neighbours may also have large degree. This is the mechanism of Henning, van der Hofstad and Litvak [17, Section 3]. We adapt it to every tail index.

Theorem 11.2 (Degree-tail transfer fails for every index $\alpha > 1$)**.** *Fix $\alpha > 1$ and put $\beta = \alpha + 2$. For $L \ge 4$ take $a_{m,L} = \lfloor L^\beta/m^\beta \rfloor$ copies of K_m for $3 \le m \le L$. Order the cliques by nondecreasing size as $C_1, \dots, C_K$, choose one portal vertex in each, and join consecutive portals by single edges. The resulting connected graphs G_L have minimum degree two and satisfy:*

1. *$|E(G_L)|/n_L \to [(S_{\beta-2} - S_{\beta-1})/2 + S_\beta]/S_{\beta-1}$, where $S_r = \sum_{m \ge 3} m^{-r}$;*
2. *the degree of a uniform vertex converges in law to D, with $\mathbb{P}(D = d) = S_{\beta-1}^{-1}[d(d+1)^{-\beta} + \mathbf{1}_{\{d \ge 4\}}(d-1)^{-\beta}]$ for integers $d \ge 2$ (and zero otherwise), and $\mathbb{P}(D > x) \sim x^{-\alpha}/(\alpha S_{\beta-1})$;*
3. *for every ε, $n_L\pi_\infty(U_L)$ converges in law to the distribution placing mass $(m-1)m^{-\beta}/S_{\beta-1}$ at $x_m = 1 - \frac{2\varepsilon}{(1+\varepsilon)(m^2-1)}$ and mass $m^{-\beta}/S_{\beta-1}$ at $y_m = 1 + \frac{2\varepsilon}{(1+\varepsilon)(m+1)}$, for $m \ge 3$; and $n_L\pi_\infty(i) \le (1 + 5\varepsilon/3)/(1 + \varepsilon) < 4/3$ for every L and i.*

Proof. Since $\beta > 3$, the clique count, vertex count and edge count, divided by L^β, converge to S_β, $S_{\beta-1}$ and $(S_{\beta-2} - S_{\beta-1})/2 + S_\beta$. The rounding errors are $O(L)$, $O(L^2)$ and $O(L^3) = o(L^\beta)$. A nonportal vertex of K_m has degree $m - 1$. A portal has degree $m - 1 + b$, $b \in \{1, 2\}$, with $b = 2$ except at the two ends of the chain. Counting nonportals and internal portals gives the degree law. Its first term is $\sim d^{1-\beta}$, and $\sum_{d>x} d^{1-\beta} \sim x^{2-\beta}/(\beta - 2)$.

For the score law, discard the cliques with indices in

$$\mathcal{E}_L = \{1, 2, K-1, K\} \cup \bigcup_{j:\, |C_j| \ne |C_{j+1}|} \{j, j+1\}.$$

There are at most L size changes and every clique has at most L vertices, so at most $(4 + 2L)L = O(L^2) = o(n_L)$ vertices are discarded. A surviving clique C_j of size m has neighbouring cliques $C_{j\pm1}$ of the same size that are not endpoints, so both external neighbours of its portal have degree exactly $m + 1$. Its nonportal vertices then have $h = (m-2)/(m-1) + 1/(m+1)$ and its portal has $h = 1 + 2/(m+1)$, which give x_m and y_m by (11). (Without discarding the neighbours of the end cliques this fails: for $L = 4$, $\beta = 4$, the chain K_3–K_3–K_3–K_4 has portal score $43/36$ in the second clique instead of $y_3 = 42/36$ at $\varepsilon = 1/2$.) The surviving vertices follow the stated law, so the limit holds. For the bound: nonportal vertices have $h \le 1$; a portal has $h \le 1 + 2/3$, since its external neighbours are portals of degree at least three. □

The theorem does not contradict positive results under extra assumptions. In a configuration model with i.i.d. positive degrees of finite mean, the neighbour degrees are asymptotically independent size-biased variables D^* with $\mathbb{E}[1/D^*] = 1/\mathbb{E}[D]$ (with isolated vertices allowed this becomes $\mathbb{P}(D > 0)/\mathbb{E}[D]$). This suggests that $n\pi_\infty(U)$ converges in law to $Z = (1+\varepsilon\sum_{k\le D} 1/D_k^*)/(1+\varepsilon)$. For this limiting variable the tail transfer is elementary: the summands are bounded and i.i.d., so concentration of $\sum_{k\le D} 1/D_k^*$ around $D/\mathbb{E}[D]$ and regular variation give $\mathbb{P}(Z > x) \sim [\varepsilon/((1+\varepsilon)\mathbb{E}[D])]^\alpha\, \mathbb{P}(D > x)$. What we have not proved is the graph-level statement: a precise model, conditioning and degree hypotheses, and local convergence of this radius-two functional. We state that as an open problem. Together with Proposition 11.1, it would make the picture two-sided: neutral mixing transfers the tail, assortative mixing can destroy it, and nothing can make the score tail heavier than the degree tail.

12 Verification, literature boundary and scope

Code. `code/verify_all.py OUTDIR` replays every exact certificate and the census, and writes deterministic JSON. It uses Python fractions, python-flint integer polynomials and sympy exact fields over $\mathbb{Q}(\mu, \varepsilon)$. Specifically, it:

- checks the house identities of Theorems 3.1–4.1 as polynomials, and the threshold brackets, by exact signs;
- checks the house formula against 25 independent full-state rational solves, and checks two- and three-sheet covers;
- derives the P_3, paw, gem and diamond responses and all six order-four responses symbolically, and certifies every sign claim in Propositions 5.1 and 5.3 and Corollary 5.4 by exact Sturm root counts;
- runs the census of Section 7, after root-counter regressions (two synthetic responses with roots at $\mu = 0$ and inside $(0, 1)$, Atlas graph 399 at $\varepsilon = 1/2$, and all subsets of a known root set);
- checks Proposition 6.2 exactly at 3985 vertices, together with the two-step comparison of Section 6;
- checks the tree certificate of Proposition 5.5, the spread certificates of Propositions 8.1 and 8.2, the Les Misérables reversal and the finite rank-reversal search;
- constructs clique chains for $\beta = 4$ ($L = 4, 5, 8, 12, 16$) and $\beta = 5$, checking the generic scores on every surviving clique and the regression value $43/36$.

The numerical prevalence study is a separate script. Graph-cover preservation, the period-class argument and the asymptotic statements are proved in prose. No proof assistant was used.

Literature. We inspected arXiv:2211.13353v2 [2] in full. The journal version [1] was inaccessible, so its conjecture wording is unverified. We also inspected the full papers [3, 5], the relevant parts of [10, 21, 6], and Section 3 of [17]. Related work on parameter dependence [9, 12], higher-order PageRank [13, 7] and non-backtracking walks [16, 4, 19] frames the setting but does not address coordinatewise monotonicity in μ. Within this search we found no prior counterexample or census; no exhaustive priority claim is made.

13 Conclusions and open problems

The coordinatewise monotonicity conjectured by Glover et al. fails, and it fails often. With minimum degree two the smallest failures have five vertices at every damping value. With leaves allowed the smallest order is four for $\varepsilon < \varepsilon_1$ or $\varepsilon > \varepsilon_2$ and five for $\varepsilon_1 \le \varepsilon \le \varepsilon_2$, and even a

five-vertex tree fails. Exhaustive exact computation at five damping values shows failure in most seven-vertex graphs. On standard networks between a tenth and two fifths of vertices are affected (numerically). The failures are small in size, and rankings are largely stable, so the practical lesson is specific: one cannot infer the side of the non-backtracking score from ordinary and infinite-backtracking scores alone. Proposition 6.2 and the tests of Corollary 6.3 give a cheap partial substitute. Both readings of the spread clause are false, even for graphs whose extreme vertices are unique, and the degree-tail conjecture fails for every tail index above one.

Open problems: (1) a structural classification of graphs, or vertices, with monotone coordinates for all damping values; trees are not such a class (Proposition 5.5); (2) a graph-level proof of the configuration-model tail transfer of Section 11; (3) a structural condition under which the reversed spread clause holds, given that uniqueness of the extreme vertices is not sufficient (Proposition 8.2); (4) certified extension of the census to order eight, where nauty enumeration makes it feasible, and to the whole damping interval; (5) the effect of the node2vec in–out parameter, which this paper fixes at one.

Data and code availability. All code (MIT licence) and results are in the accompanying package, archived at https://doi.org/10.5281/zenodo.23580381 and available at https://github.com/ipitchford/pagerank-backtracking-nonmonotone (tag `v1.0.0-candidate`); the text is released under CC0-1.0.

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